When Pooling Cannot Save a Test

A guided proof for adaptive group testing with a known defective count

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Abstract

Suppose we know that exactly d of n items are defective. A test of a group of items tells us only whether the group contains at least one defective. We may choose each new group after seeing the earlier answers. Can this freedom guarantee fewer tests than examining items one at a time?

This exposition proves that the answer is no when $n \geq 2^{31}$ and $d/n \geq 369056/1000000$: the optimal worst-case number is exactly $n-1$. We begin with small examples, explain decision trees, and develop all the probability tools used in the proof. In particular, entropy, relative entropy, conditioning on an exact count, and an artificial comparison experiment are introduced from finite-sum definitions. A structural argument about legal pooled tests then connects a probability estimate near density one quarter to the stated theorem. Every mathematical ingredient needed for the conclusion is proved here. The large numerical constants are accompanied by exact certificates, with the longer arithmetic placed in an appendix.

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1 The problem: what can a pooled test tell us?

1.1 Items, defective sets, and answers

Imagine five labeled items, numbered $1, 2, 3, 4, 5$. Each item is either defective or nondefective. The set of defective labels might be

$$D = \{2, 5\}.$$

The label *defective* simply names one of two possible statuses. No statistical model is initially imposed on those statuses. There is a fixed unknown set, and we must identify it exactly.

A *pool* is a subset of the labels. Testing the pool $S = \{1, 2, 3\}$ gives a positive answer for this D, because label 2 lies in both sets. Testing $\{1, 3\}$ gives a negative answer, because neither label is defective. A negative answer certifies that every item in the tested pool is nondefective. A positive answer says only that at least one item in it is defective. It does not say which one, or how many.

Write $[n] = \{1, \ldots, n\}$. Formally, the answer to pool $S \subseteq [n]$ on defective set $D \subseteq [n]$ is

$$\mathrm{OR}_D(S) = \begin{cases} 0, & S \cap D = \varnothing, \\ 1, & S \cap D \neq \varnothing. \end{cases}$$

This is called a pooled-OR test: the answer is the logical OR of the individual defective indicators in the pool. We call 0 negative and 1 positive.

Throughout this document the tests are noiseless, so an answer is always correct in an actual execution of the testing algorithm. The algorithm is deterministic: a given history of answers determines its next query. It is fully adaptive: that query may depend on the entire history, and may be any subset of the labels.

1.2 An adaptive strategy must handle all its branches

Example 1.1. There is exactly one defective among four items. First test $\{1, 2\}$. If the answer is positive, test $\{1\}$: the defective is 1 on a positive answer and 2 on a negative one. If the first answer is negative, test $\{3\}$: the defective is 3 on a positive answer and 4 on a negative one. Thus two tests suffice on every possible input.

The second test depends on the first answer. If the actual defective is 3, only the negative-then-positive branch is executed. Nevertheless, a correct algorithm must specify successful behavior on the other three inputs as well. Finding a short explanation for one particular defective set is easier than designing a strategy that identifies every possible defective set within a uniform test limit. Our lower bound concerns the latter requirement.

1.3 What is known in advance?

The labels and the exact integer d are known. The unknown input is one of the d-element subsets of $[n]$. We write $\binom{[n]}{d}$ for this family of sets, and $\binom{n}{d}$ for its number of members. The promise is always $0 < d < n$. An algorithm may use the exact count at any stage. Once it has identified all d defectives, for instance, it knows every other label is nondefective without testing them.

Knowing the count is different from knowing only an approximate defective fraction. If four tested items have revealed one defective and the total is known to be two among five, then the fifth item is defective. Without the exact total, that conclusion would be unjustified.

Definition 1.2. The *worst-case number of tests* of an algorithm is the largest number it uses on any $D \in \binom{[n]}{d}$. The number $M(d,n)$ is the minimum of this worst-case number over all correct deterministic adaptive pooled-OR algorithms.

The order matters: first take the worst input for a fixed algorithm, then choose the best algorithm. Different algorithms may have different worst inputs.

Proposition 1.3. *For $0 < d < n$, $M(d,n) \le n-1$.*

Proof. Test labels $1, \dots, n-1$ individually. If they contain r defectives, the last item must have defective indicator $d-r$. For a valid input this indicator is either zero or one, so its status is determined. This is a correct algorithm using at most $n-1$ tests. □

Sometimes this strategy stops earlier, but its bound on all inputs is $n-1$. A pooled strategy would improve on it in the worst case only if it guaranteed at most $n-2$ tests for *every* valid input.

1.4 The theorem we will prove

Theorem 1.4 (Individual testing in a dense regime)**.** *Let $c_\star = 369056/1000000$. For every pair of integers n, d satisfying*

$$n \ge 2^{31}, \qquad 0 < d < n, \qquad \frac{d}{n} \ge c_\star,$$

one has

$$\boxed{M(d,n) = n-1.}$$

In ordinary language: for every population in the stated size range, if at least the fraction 369056/1000000 is defective, no legal adaptive pooling strategy can guarantee a saving over individual testing. Some inputs might be identified very quickly; the theorem says that every correct strategy has at least one input requiring $n-1$ or more tests.

Proposition 1.3 already gives the upper bound. The work lies in the opposite direction, $M(d,n) \ge n-1$. Thus this is a lower-bound proof for the number of tests. It must account for every possible strategy, including strategies whose pools have complicated sizes and overlaps.

The number $2^{31} = 2147483648$ is an explicit sufficient cutoff in the proof. We do not claim it is necessary. The theorem has no further exceptions within its range, but makes no corresponding general assertion for smaller populations.

2 Decision trees and the first counting bound

2.1 Putting a strategy on paper

A deterministic adaptive algorithm can be drawn as a rooted binary tree. An internal node v contains a pool S_v. Its two outgoing edges have labels 0 and 1. A test answer tells us which edge to take. A leaf ℓ is a stopping point with an output defective set. Correctness means that every valid input reaches a leaf whose output is that input.

A *history* or *prefix transcript* τ is the finite string of answers seen so far. In a fixed tree it determines a node, as long as execution has not already stopped. A *terminal transcript* y is the full

answer string leading to a leaf. We distinguish the string y from the leaf ℓ, although the tree gives a one-to-one correspondence between them. The depth of a leaf is its number of preceding tests.

Here is Example 1.1 as a table:

Defective set	Queries along its path	Answers
$\{1\}$	$\{1,2\}$, then $\{1\}$	11
$\{2\}$	$\{1,2\}$, then $\{1\}$	10
$\{3\}$	$\{1,2\}$, then $\{3\}$	01
$\{4\}$	$\{1,2\}$, then $\{3\}$	00

The same first query appears in all rows. The second query is a function of the first answer, not a function of an unknown input that the algorithm has somehow been told.

We only need finite trees. A strategy with a finite worst-case test bound can be restricted to its valid-input paths, giving a finite tree. An algorithm without a finite worst-case bound cannot improve the finite strategy of Proposition 1.3.

2.2 Counting leaves

Proposition 2.1. *Every correct tree for $\binom{[n]}{d}$ has worst-case depth at least $\lceil \log_2 \binom{n}{d} \rceil$.*

Proof. Two different valid inputs cannot terminate at the same leaf, because the leaf has only one output. Thus at least $\binom{n}{d}$ valid leaves are needed. A binary tree of depth at most s has at most 2^s leaves. This follows by induction: at depth zero there is at most one; each of two child subtrees has at most 2^{s-1} leaves. Therefore $2^s \geq \binom{n}{d}$. Taking logarithms and using that s is an integer gives the assertion. □

This bound uses only the two possible answers per test. It would hold even if arbitrary yes/no questions about the input were allowed. Such questions can be more flexible than pooled tests.

For instance, when $d = n-1$, there are only n possibilities: the locations of the one nondefective item. Counting gives only $\lceil \log_2 n \rceil$. Yet every pool of size at least two is always positive, so it reveals nothing. We will prove in Section 10 that the actual optimum is $n - 1$. This shows why counting alone cannot establish the desired theorem.

2.3 A pooled test cannot make an arbitrary split

Suppose $n = 4, d = 2$, giving six candidate inputs. A negative answer to S leaves exactly the candidates contained in $[4] \setminus S$. If $|S| = k$, their number is $\binom{4-k}{2}$, interpreted as zero if $4 - k < 2$. Depending on k, this number is $6, 3, 1$, or 0, never 2.

An unrestricted binary question could ask whether the input belongs to a chosen two-element collection of candidates. A pooled test cannot produce that negative branch on this complete family. Later the candidate family is more complicated, but an answer must still be determined by whether the pool intersects the defective set. Replacing a legal query by a conveniently balanced arbitrary binary partition would change the problem.

3 How the proof fits together

The proof has two main parts. One is probabilistic and measures what a legal tree can accomplish near defective density 1/4. The other is structural and relates optimal test counts on different population sizes. Neither part alone proves the theorem.

For the probabilistic part we choose a uniformly random valid defective set solely as a tool for analyzing a fixed deterministic tree. Let

$$H = \log_2 \binom{n}{d}, \qquad L = \text{its expected number of tests.}$$

We will explain entropy and prove $L \geq H$. The difference $\Gamma = L - H$ measures how much expected test count exceeds this basic information requirement.

The important estimate is stronger:

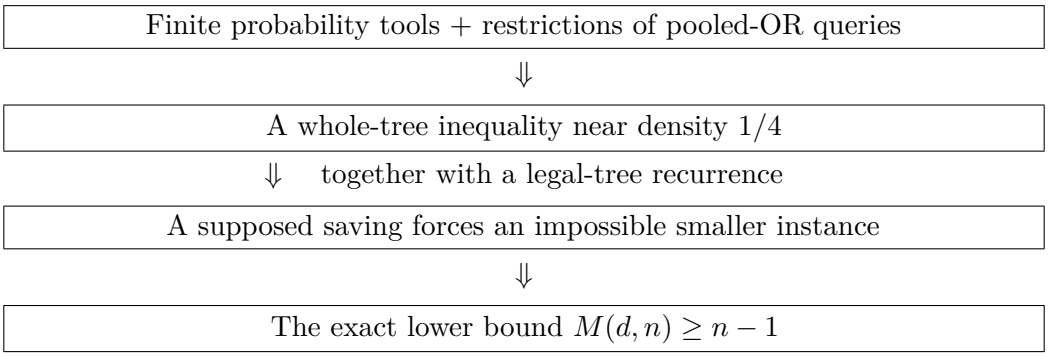
$$6615(L - H) \geq n - H - 67229K, \qquad K = nh(d/n) - H, \tag{3.1}$$

for $249/1000 \leq d/n \leq 251/1000$. Here h is the entropy of one binary outcome, defined below. The quantity K is a cost for comparing exact-count inputs with independent inputs. We will prove $0 \leq K \leq \log_2(n+1)$. Thus it is a logarithmic correction, rather than a new linear term.

Why introduce independent inputs? Independence makes probabilities of all-good pools easy to calculate. But the real inputs have an exact count and are not independent. Relative entropy will keep track of the difference over an entire tree. We also introduce a second comparison experiment that answers large pools by fair coin flips. This artificial experiment is not a testing algorithm for the original problem. It leaves certain input coordinates independent after a transcript, allowing us to calculate how much uncertainty remains.

The structural part proves a recurrence for M. In a hypothetical instance where some strategy saves a test, the recurrence forces smaller instances with increasingly strong savings. Following this implication eventually reaches a density just below 1/4, where (3.1) applies. The forced saving is then too large to be compatible with that inequality.

This explains why an estimate at density 1/4 matters for a theorem whose initial density is at least $c_\star$. The recurrence connects the two regimes; the initial problem is not assumed to have density 1/4.



All probability spaces below are finite. Randomness is introduced to prove a lower bound, not to allow randomized testing strategies. Every expectation relevant to the real algorithm averages over valid exact-d inputs.

The notation is introduced gradually. A fixed input is D, a random input is X, a query is S_v, a node is v, a prefix transcript is τ, a terminal transcript is y, and a leaf is ℓ. The random terminal transcript is Y. We write P_X for an input law, P_Y for a transcript law, and $P_{X,Y}$ for their joint law. This subscript convention also applies to the comparison laws. A full notation guide appears in Appendix B; each symbol is explained in the text before it is used.

4 Entropy: uncertainty measured by a finite sum

4.1 Why logarithms appear

An outcome of probability $1/8$ has the same probability as three specified successive fair binary choices. This motivates assigning it information content $-\log_2(1/8) = 3$. More generally, assign an outcome of probability $u > 0$ information content $-\log_2 u$. The identity

$$-\log_2(uv) = -\log_2 u - \log_2 v$$

lets us add information along conditional choices. This logarithmic identity, rather than a coding theorem, is what we will use. From now on log means $\log_2$, and ln means natural logarithm. The unit is a bit.

Definition 4.1. For a law μ on a finite set Ω, its *entropy* is

$$\mathsf{H}(\mu) = -\sum_{z\in\Omega} \mu(z)\log\mu(z).$$

We use $0\log 0 = 0$, as justified by the limit when the probability tends to zero.

Entropy is the average information content under μ. It is nonnegative because $-\log\mu(z) \geq 0$ for every outcome of positive probability. For a uniform law on N outcomes,

$$\mathsf{H}(\mu) = -N\frac{1}{N}\log\frac{1}{N} = \log N.$$

A certain outcome has entropy zero. Information content of an individual outcome and entropy of a law are distinct: the latter averages the former.

4.2 One binary outcome

The entropy of two answers, with probabilities u and $1-u$, is

$$h(u) = -u\log u - (1-u)\log(1-u), \qquad 0 \leq u \leq 1.$$

It equals one at $u = 1/2$, and zero at $u = 0$ or 1. For example, direct substitution gives $h(1/4) = 2 - \frac{3}{4}\log 3$. Calculus quantifies the fact that biased answers have less entropy:

$$h'(u) = \log\frac{1-u}{u}, \qquad h''(u) = -\frac{1}{(\ln 2)u(1-u)} < 0 \quad (0 < u < 1).$$

The derivative is positive below $1/2$ and negative above it. Together with the endpoint values, this proves

$$0 \leq h(u) \leq 1, \qquad h(u) = 1 \iff u = 1/2. \tag{4.1}$$

4.3 Conditional entropy and its chain rule

For finite random variables U, V, their *joint law* records the probabilities of pairs: $\mu_{U,V}(u,v) = \Pr(U = u, V = v)$. Their *marginal laws* record the probabilities of just one variable; for example, $\mu_V(v) = \sum_u \mu_{U,V}(u,v)$. When $\mu_V(v) > 0$, the conditional law is

$$\mu_{U|V=v}(u) = \frac{\mu_{U,V}(u,v)}{\mu_V(v)}.$$

Conditioning values of zero probability are omitted from conditional averages. With this notation, define

$$\mathsf{H}_\mu(U \mid V) = \sum_{v:\mu_V(v)>0} \mu_V(v)\mathsf{H}(\mu_{U|V=v}).$$

This is an average of entropies of conditional laws, weighted by how often each conditioning value occurs.

Lemma 4.2 (Entropy chain rule)**.** $\mathsf{H}(\mu_{U,V}) = \mathsf{H}(\mu_V) + \mathsf{H}_\mu(U \mid V)$.

Proof. For every pair of positive joint probability, $\mu_{U,V}(u,v) = \mu_V(v)\mu_{U|V=v}(u)$. Take negative logarithms and average:

$$\begin{aligned} -\sum_{u,v} \mu_{U,V}(u,v) \log \mu_{U,V}(u,v) = &-\sum_{u,v} \mu_{U,V}(u,v) \log \mu_V(v) \\ &-\sum_{u,v} \mu_{U,V}(u,v) \log \mu_{U|V=v}(u). \end{aligned}$$

Summing the first term over u gives $\mathsf{H}(\mu_V)$. Factoring $\mu_V(v)$ out of the second for each v gives the defined conditional entropy. □

Thus learning a pair can be separated into learning V, then learning what remains uncertain about U when V is known. Independence is not assumed.

5 Relative entropy: comparing two descriptions

5.1 Definition, meaning, and support

We will run the same tree under different probability models. The quantity that compares these laws must involve two distributions.

Definition 5.1. For finite laws μ, ν on the same set, their *relative entropy*, or Kullback–Leibler divergence, is

$$\mathrm{KL}(\mu\|\nu) = \sum_{z:\mu(z)>0} \mu(z) \log \frac{\mu(z)}{\nu(z)}.$$

If $\nu(z) = 0 < \mu(z)$ for some z, its value is $+\infty$. If every outcome of positive μ-mass has positive ν-mass, we say that μ is absolutely continuous with respect to ν.

The logarithm measures how differently the two laws weight an outcome. The average is under the *first* law μ. Individual logarithms can be negative, although the sum cannot be. Entropy measures uncertainty within one law. Relative entropy measures discrepancy between two laws,

with an indicated order; it is not a symmetric distance. For example, $\mu = (1, 0)$, $\nu = (1/2, 1/2)$ give $\mathrm{KL}(\mu\|\nu) = 1$ but $\mathrm{KL}(\nu\|\mu) = +\infty$. If $\mu = \nu$, their divergence is zero. If μ is a point mass at z, then

$$\mathrm{KL}(\mu\|\nu) = \log \frac{1}{\nu(z)}. \tag{5.1}$$

Lemma 5.2 (Nonnegativity)**.** $\mathrm{KL}(\mu\|\nu) \geq 0$.

Proof. Calculus gives $-\ln s \geq 1 - s$ for $s > 0$: the difference has derivative $1 - 1/s$ and minimum zero at $s = 1$. For finite divergence, apply this with $s = \nu(z)/\mu(z)$:

$$\begin{aligned}(\ln 2)\,\mathrm{KL}(\mu\|\nu) &\geq \sum_{\mu(z)>0} (\mu(z) - \nu(z)) \\ &= 1 - \sum_{\mu(z)>0} \nu(z) \geq 0.\end{aligned}$$

Infinite divergence is also nonnegative. □

Comparison with the uniform law on N outcomes gives $\mathrm{KL}(\mu\|\text{uniform}) = \log N - \mathsf{H}(\mu)$. Thus entropy on those outcomes is at most $\log N$. This statement is a consequence of the finite-sum proof, without appealing to a separate information-theoretic theorem.

5.2 The chain rule and the direction of comparison

Lemma 5.3 (Relative-entropy chain rule)**.** *For finite joint laws with $\mu_{U,V}$ absolutely continuous with respect to $\nu_{U,V}$,*

$$\mathrm{KL}(\mu_{U,V}\|\nu_{U,V}) = \mathrm{KL}(\mu_V\|\nu_V) + \sum_{v:\mu_V(v)>0} \mu_V(v)\,\mathrm{KL}(\mu_{U|V=v}\|\nu_{U|V=v}). \tag{5.2}$$

In particular, the marginal divergence is at most the joint divergence.

Proof. On positive μ-mass pairs, factor the likelihood ratio:

$$\frac{\mu_{U,V}(u,v)}{\nu_{U,V}(u,v)} = \frac{\mu_V(v)}{\nu_V(v)} \frac{\mu_{U|V=v}(u)}{\nu_{U|V=v}(u)}.$$

Take logarithms and average under μ. The first term sums to the marginal divergence. In the second, average first under the μ-conditional law for each v, then with weight $\mu_V(v)$. This is the displayed identity. Nonnegativity of each conditional divergence proves the last assertion. □

We will use U for the hidden input and V for its transcript. The identity separates discrepancy visible in the transcript from discrepancy remaining in the conditional input law. If the joint divergence is known and the conditional term has a lower bound, subtracting that lower bound gives an *upper* bound on transcript divergence. We will write the identity again when its terms have concrete meanings.

If $V = f(U)$ is the same deterministic function under both laws, the joint masses are the input masses attached to $(u, f(u))$. Therefore $\mathrm{KL}(\mu_{U,V}\|\nu_{U,V}) = \mathrm{KL}(\mu_U\|\nu_U)$. The chain rule shows that applying the function cannot increase divergence. Truthful execution of a fixed tree is one such function.

6 Adding information along a decision tree

6.1 Why reach probabilities are the weights

Fix a finite binary tree. A law μ_Y on terminal transcripts induces a reach probability $\mu(v)$ at each internal node. For $\mu(v) > 0$, let π_v be its conditional negative-answer probability. Nodes of zero reach probability contribute zero. Let T be the random number of tests.

Lemma 6.1 (Tree decompositions)**.** *One has*

$$\mathbb{E}_\mu T = \sum_v \mu(v), \tag{6.1}$$

$$\mathsf{H}(\mu_Y) = \sum_v \mu(v) h(\pi_v). \tag{6.2}$$

For another transcript law ν_Y, with conditional negative probabilities ρ_v, and with μ_Y absolutely continuous with respect to ν_Y,

$$\mathrm{KL}(\mu_Y \| \nu_Y) = \sum_v \mu(v)\, \mathrm{kl}(\pi_v \| \rho_v), \tag{6.3}$$

where

$$\mathrm{kl}(u\|w) = u \log \frac{u}{w} + (1-u) \log \frac{1-u}{1-w}$$

is the relative entropy of two binary laws, with the same zero-probability and infinite-divergence conventions as above.

Proof. On each path, T is the sum of the indicators of visiting internal nodes. Taking expectations proves (6.1).

The probability of a terminal transcript is the product of the conditional probabilities of its edges. Expand the negative logarithm of that product and average over terminal transcripts. A node v appears only on paths that reach it, with total probability $\mu(v)$. Conditional on reaching it, its average contribution is $-\pi_v \log \pi_v - (1-\pi_v)\log(1-\pi_v) = h(\pi_v)$. This proves (6.2). For (6.3), expand the logarithm of the ratio of the two products. The conditional average at a reached node is $\mathrm{kl}(\pi_v\|\rho_v)$. All sums are finite, so their order can be changed without a convergence issue. □

The reach weight is indispensable. A test available only after a rare history contributes only when that history occurs. An unweighted sum over nodes would pretend that every branch is traversed in one run.

Example 6.2. Give the leaves of a binary tree depths $1, 2, 2$ and probabilities $1/2, 1/4, 1/4$. The root is reached with probability one and splits fairly. The other internal node is reached with probability $1/2$ and also splits fairly. Both expected depth and entropy are $1 + (1/2)\cdot 1 = 3/2$. This illustrates a finite-tree identity; it does not assert that arbitrary probability splits are realizable by pooled queries.

6.2 Expected depth beyond the counting requirement

In a correct tree on $\binom{[n]}{d}$, choose the input uniformly. Distinct valid inputs have distinct terminal transcripts, so the transcript law is uniform on $\binom{n}{d}$ outcomes. Its entropy is $H = \log \binom{n}{d}$. Let P denote this truthful exact-count experiment, and put

$$L = \mathbb{E}_P T, \qquad \Gamma = L - H.$$

The tree identities give

$$\Gamma = \sum_v P(v)\big(1 - h(\pi_v)\big) \geq 0, \qquad \pi_v = P(\text{answer } 0 \mid \text{reach } v). \tag{6.4}$$

Thus Γ is both the expected test count in excess of H and the visit-weighted sum of the amounts by which answers fall short of one bit of entropy.

Averages satisfy $L \leq \max_D T(D)$, so a lower bound on L is also a lower bound on that maximum. But a modest improvement over H may still be below $n - 1$. The structural part of the proof will be needed for the exact conclusion.

6.3 A weighted comparison for one answer

Suppose the negative probability is π in the actual experiment and ρ in a comparison experiment, with ρ far from $1/2$. If π is also far from $1/2$, the deficiency $1 - h(\pi)$ is substantial. If π is near $1/2$, it differs from ρ, producing relative entropy. The following lemma quantifies this alternative.

Lemma 6.3 (Weighted binary inequality)**.** *For $\omega > 0$ and $\pi, \rho \in [0, 1]$,*

$$\omega\big(1 - h(\pi)\big) + \mathrm{kl}(\pi \| \rho) \geq \frac{2\omega}{\omega + 1} \frac{(\rho - 1/2)^2}{\ln 2}. \tag{6.5}$$

Proof. For $0 < \rho < 1$, let $f(u) = \mathrm{kl}(u \| \rho)$. Differentiation gives

$$f(\rho) = f'(\rho) = 0, \qquad f''(u) = \frac{1}{(\ln 2) u(1 - u)} \geq \frac{4}{\ln 2}.$$

Integrate twice from ρ to u, reversing directions when $u < \rho$, to get $f(u) \geq 2(u - \rho)^2 / \ln 2$. Apply this also with $\rho = 1/2$, noting $\mathrm{kl}(u \| 1/2) = 1 - h(u)$. The left side of (6.5) is then at least

$$\frac{2}{\ln 2} \left[\omega(\pi - 1/2)^2 + (\pi - \rho)^2\right].$$

Complete the square:

$$\omega(\pi - 1/2)^2 + (\pi - \rho)^2 = (\omega + 1) \left(\pi - \frac{\omega/2 + \rho}{\omega + 1}\right)^2 + \frac{\omega}{\omega + 1}(\rho - 1/2)^2.$$

Discarding the square proves the result. Continuity handles finite endpoint cases; an infinite relative entropy satisfies the inequality as well. □

The weight ω balances the two terms when summing over node classes. It changes neither the algorithm nor a probability law. We will use $1/10$ for large pools and 64 for certain pairs. Their role is to make the final coefficient calculation work, with every numerical comparison checked exactly.

7 Independent inputs and the cost of knowing the exact count

The real problem fixes the number of defectives. This section introduces an independent input law for comparison and quantifies the cost of conditioning it on the required count. At first we discuss input laws only; executions on other counts will be defined in the next section.

7.1 Two input laws on the same finite set

Let $X \subseteq [n]$ be a random subset, with indicators $X_i = \mathbf{1}\{i \in X\}$. Set $p = d/n$ and $q = 1-p$. There are two ways to choose X:

- Under P_X, choose uniformly among the d-element subsets. Thus $P_X(D) = 1/\binom{n}{d}$ for $|D| = d$, and zero otherwise.
- Under Q_X, choose the indicators independently, each equal to one with probability p. Thus

$$Q_X(D) = p^{|D|}q^{n-|D|} \qquad (D \subseteq [n]).$$

The second is a Bernoulli product law. Its expected defective count is $np = d$, but its count is not fixed. Having the same mean as the promised count does not make it the original input law.

Under P_X, each item individually has defective probability p, but items are generally dependent. With exactly one defective among four,

$$P_X(X_2 = 1) = \frac{1}{4}, \qquad P_X(X_2 = 1 \mid X_1 = 1) = 0.$$

Under the independent law with $p = 1/4$, the latter probability would still be $1/4$.

All d-subsets have the same Q_X-mass $p^d q^{n-d}$. Consequently

$$P_X = Q_X(\,\cdot \mid |X| = d). \tag{7.1}$$

This is an identity, not an approximation. For independent subsets,

$$\mathsf{H}(Q_X) = -\mathbb{E}_{Q_X}[|X| \log p + (n - |X|) \log q] = nh(p),$$

because $\mathbb{E}_{Q_X}|X| = np$. Since $\mathsf{H}(P_X) = H = \log \binom{n}{d}$, define

$$K = nh(p) - H. \tag{7.2}$$

The next calculation explains why this particular difference also has a divergence interpretation.

Lemma 7.1 (Exact-count comparison)**.** *For the laws just defined,*

$$\mathrm{KL}(P_X \| Q_X) = -\log Q_X(|X| = d) = K, \qquad 0 \le K \le \log(n+1). \tag{7.3}$$

Proof. For every valid input D,

$$\frac{P_X(D)}{Q_X(D)} = \frac{1}{\binom{n}{d}p^d q^{n-d}} = \frac{1}{Q_X(|X| = d)}.$$

The logarithm of this ratio is constant on the support of P_X. Its average equals that same logarithm, namely $-H - d\log p - (n-d)\log q = nh(p) - H$.

For $0 \le k \le n$, put $b_k = Q_X(|X| = k) = \binom{n}{k}p^k q^{n-k}$. For $k < n$,

$$\frac{b_{k+1}}{b_k} = \frac{n-k}{k+1}\frac{d}{n-d}.$$

This ratio decreases with k. At $k = d-1$ it equals $(n-d+1)/(n-d) > 1$; at $k = d$ it equals $d/(d+1) < 1$. Hence b_d is a largest one of $b_0, \ldots, b_n$. These $n+1$ numbers sum to one, so $1/(n+1) \le b_d \le 1$. Taking negative logarithms proves the bounds. □

Example 7.2. For $n = 4, d = 1, p = 1/4$,

$$Q_X(|X| = 1) = 4\left(\frac{1}{4}\right)\left(\frac{3}{4}\right)^3 = \frac{27}{64}.$$

Conditioning gives the uniform law on the four defective labels. The comparison cost is $K = \log(64/27)$. The equality of costs across valid inputs is important; it does not assert a small cost for every later conditioning event.

In particular, $K \le \log(n+1)$ is one bound for the entire input-law comparison. It is not a new bound that may be spent independently at each node. The tree chain rule will keep the accumulated discrepancy within this single total.

7.2 Increasing and decreasing events

Identify subsets with their 0-1 vectors. A function is *increasing* if changing any coordinate from zero to one cannot decrease its value. An event is increasing if its indicator is increasing. It is *decreasing* if adding defectives cannot turn a false event into a true one.

For fixed S, the event $X \cap S \neq \varnothing$ is increasing, and $X \cap S = \varnothing$ is decreasing. An intersection of positive-test events is increasing too: adding defectives preserves all the positive answers.

Lemma 7.3 (Association for independent coordinates)**.** *Under a product law on finitely many independent Bernoulli coordinates, increasing real functions f, g satisfy $\mathbb{E}(fg) \ge (\mathbb{E}f)(\mathbb{E}g)$. If I is increasing with positive probability and J_0 is decreasing, then*

$$\Pr(J_0 \mid I) \le \Pr(J_0). \tag{7.4}$$

Proof. Define $\operatorname{Cov}(f, g) = \mathbb{E}(fg) - (\mathbb{E}f)(\mathbb{E}g)$. Induct on the number of coordinates. With no coordinates the functions are constant and covariance is zero. Let Z be the last coordinate. Conditional on Z, the other coordinates remain independent, so the induction hypothesis gives $\operatorname{Cov}(f, g \mid Z) \ge 0$. Put $f_i = \mathbb{E}(f \mid Z = i)$, $g_i = \mathbb{E}(g \mid Z = i)$. The other-coordinate law is the same at $i = 0$ and 1, so pointwise increase gives $f_1 \ge f_0$, $g_1 \ge g_0$. If $\Pr(Z = 1) = u$, expansion gives

$$\operatorname{Cov}(f_Z, g_Z) = u(1-u)(f_1 - f_0)(g_1 - g_0) \ge 0.$$

Add and subtract $\mathbb{E}[\mathbb{E}(f \mid Z)\mathbb{E}(g \mid Z)]$ to obtain

$$\operatorname{Cov}(f, g) = \mathbb{E}\operatorname{Cov}(f, g \mid Z) + \operatorname{Cov}(\mathbb{E}(f \mid Z), \mathbb{E}(g \mid Z)).$$

Both terms are nonnegative. Constant coordinates may be omitted.

Finally, $\mathbf{1}_I$ and $-\mathbf{1}_{J_0}$ are increasing. Their nonnegative covariance gives $\Pr(I \cap J_0) \le \Pr(I)\Pr(J_0)$. Divide by $\Pr(I) > 0$. □

Coordinates need not remain independent after a positive answer. The lemma says instead that increasing constraints cannot raise the probability of a decreasing event. That inequality is what we need.

8 Preparing one tree and defining three experiments

Fix a correct deterministic tree for $\binom{[n]}{d}$. The tree is not random. We will run it under three laws. First we make its behavior unambiguous even on histories impossible under the promise.

8.1 Completing missing branches

If an answer branch was omitted because no valid input can take it, attach an immediate leaf with an arbitrary output. This changes no valid input path. It only provides an endpoint for a comparison experiment that takes a formerly omitted edge. The completed tree is finite. It need not identify inputs outside the exact-d promise; we only need all experiments to be defined.

8.2 Removing labels already declared negative

At a node v, let Z_v be the union of all original pools receiving negative answers earlier on its prefix transcript. Replace its original query S_v by

$$\widehat{S}_v = S_v \setminus Z_v. \tag{8.1}$$

This is the *normalized pool*. The set Z_v is determined by the edge labels on the path, before any probability law is chosen.

Lemma 8.1. *Normalization preserves every truthful execution on every input subset.*

Proof. At the root $Z_v = \varnothing$. Suppose the original and normalized executions have reached the same node after the same truthful answers. Every label in a previous negative pool is nondefective, so $X \cap Z_v = \varnothing$. Hence

$$X \cap S_v = \varnothing \quad \Longleftrightarrow \quad X \cap (S_v \setminus Z_v) = \varnothing.$$

The answers agree. After a negative edge, updating the union by the normalized or original pool gives the same result because $Z_v \cup (S_v \setminus Z_v) = Z_v \cup S_v$. Induction proves agreement through termination. □

This proof works for all input subsets, even those of size different from d. Empty normalized pools may be retained. From now on, pool size and label appearance refer to normalized pools. A node is *small* if $|\widehat{S}_v| \le 2$ and *large* if $|\widehat{S}_v| \ge 3$. This classification is fixed by the node and its history.

8.3 The truthful experiments

Under $P_{X,Y}$, draw X from P_X and answer all normalized queries truthfully. This is the distribution of valid executions used in the lower bound. Under $Q_{X,Y}$, draw X from Q_X and also answer truthfully. The completed tree defines the execution even when $|X| \neq d$. Under these two laws Y is the same deterministic function of X. The exact-count comparison and the chain rule therefore give

$$\mathrm{KL}(P_{X,Y} \| Q_{X,Y}) = K, \qquad \mathrm{KL}(P_Y \| Q_Y) \le K. \tag{8.2}$$

This controls accumulated discrepancy over the entire transcript.

8.4 The artificial reference experiment

The third joint law $R_{X,Y}$ is defined by:

1. Draw X from the independent law Q_X.

2. At a small node, answer with $\mathrm{OR}_X(\widehat{S}_v)$.

3. At a large node, ignore its OR value and use a new fair bit, independent of the input and all preceding bits.

4. Follow that edge and repeat until a leaf is reached.

This is a finite experiment. It could be implemented by sampling a fair bit for every large node in advance and using it if visited. The law $R_{X,Y}$ records only input and transcript; unused bits are summed out.

The next query remains adaptive, selected by the fixed tree from the answers already supplied. Normalization still follows (8.1) syntactically. An artificial negative on a large pool removes its labels from later queries, but does *not* establish that their input indicators are zero under R. A small answer is truthful for the normalized pool, which need not have the original pool's OR value under an artificial history.

Consequently a reference transcript may be inconsistent with its sampled input. This is legitimate: R is a comparison measure, not a proposed identification algorithm. The real algorithm under P still receives only truthful answers and no additional oracle.

Joint law	Input law	Answers in the normalized tree	Purpose
$P_{X,Y}$	Uniform on $\lvert X\rvert = d$	All truthful	Average over valid inputs
$Q_{X,Y}$	Independent Bernoulli-p	All truthful	Control large-pool probabilities
$R_{X,Y}$	Independent Bernoulli-p	Truthful for size at most two; fair independent bits for larger pools	Control fresh pairs and untouched labels

The Q and R input marginals agree, but their transcript and joint laws generally differ. All averaged counts in the main argument will use P, not Q or R.

8.5 A fixed-history formula

Fix a prefix τ leading to a node. Let $a(\tau)$ count its preceding large nodes. Let E_τ be the event that X gives the prescribed answers at every preceding *small normalized* pool. These pools are fixed once τ is fixed. For every $D \subseteq [n]$,

$$R(X = D,\ \text{reach } \tau) = Q_X(D)\, 2^{-a(\tau)}\, \mathbf{1}_{E_\tau}(D). \tag{8.3}$$

Indeed, each prescribed large answer has probability $1/2$; each prescribed small answer has probability one or zero according to whether D satisfies it. Multiply these factors along the fixed path, together with the input mass. The same formula holds at a terminal transcript, with reaching it meaning termination there.

Sum over D, and then divide when the reach probability is positive:

$$R(\text{reach } \tau) = 2^{-a(\tau)} Q_X(E_\tau), \qquad R(X = D \mid \text{reach } \tau) = Q_X(D \mid E_\tau).$$

In particular,

$$R_{X|\text{reach } \tau} = Q_X(\,\cdot \mid E_\tau). \tag{8.4}$$

Let V_τ be the union of the preceding small pools. The event E_τ depends only on coordinates in V_τ. For a configuration z on its complement, independence gives

$$Q_X((X_i)_{i\notin V_\tau} = z,\ E_\tau) = Q_X((X_i)_{i\notin V_\tau} = z)\, Q_X(E_\tau).$$

After division, those coordinates have their original product law under the conditional reference experiment. Thus labels absent from all preceding small pools remain mutually independent Bernoulli-p, even if they occurred in large pools. No independence within V_τ is asserted.

Example 8.2 (An artificial negative)**.** Take six labels with promised count $d = 2$, so that $p = 1/3$. Suppose the first query is $\{1, 2, 3\}$, and after answer zero the original next query is $\{2, 4, 5\}$. Its normalized version is $\{4, 5\}$. These can be initial queries of a legal strategy that finishes by individual testing on each branch.

Under R, the first answer is fair and can be zero even for the sampled input $D = \{1, 4\}$. The second normalized query then answers one truthfully. On the fixed prefix 01, the only input constraint is $X \cap \{4, 5\} \neq \varnothing$, and

$$R(X = D, \text{reach } 01) = Q_X(D)\,\frac{1}{2}\,\mathbf{1}\{D \cap \{4, 5\} \neq \varnothing\}.$$

Coordinates $1, 2, 3, 6$ remain independent Bernoulli-p conditional on this prefix. The artificial zero has not fixed $1, 2, 3$ to be nondefective. Coordinates $4, 5$, however, are constrained by a positive OR answer and need not be independent. This illustrates the experiment, not an application of the forthcoming density-1/4 inequality.

9 The whole-tree information bound near density one quarter

We now apply the tools to a fixed correct tree. Retain $p = d/n$, $q = 1 - p$, and suppose throughout this section that

$$\frac{249}{1000} \le p \le \frac{251}{1000}. \tag{9.1}$$

Near $p = 1/4$, three independent items are all good (that is, nondefective) with probability near $(3/4)^3 < 1/2$, whereas two are all good with probability near $(3/4)^2 > 1/2$. This explains the division into large pools and pairs. The actual argument must make such bounds valid after adaptive histories, not only before any tests.

Theorem 9.1 (The whole-tree inequality)**.** *Suppose* (9.1) *holds. For any finite correct deterministic adaptive tree, set*

$$L = \mathbb{E}_P T, \qquad H = \log \binom{n}{d}, \qquad K = nh(p) - H, \qquad \Gamma = L - H.$$

Then

$$\boxed{6615\Gamma \ge n - H - 67229K.} \tag{9.2}$$

The same inequality holds with $M(d, n) - H$ in place of Γ.

When the right side is positive, this forces a positive expected excess over H, and is stronger than the information bound alone. Its form will matter in the final proof: $K \le \log(n + 1)$ is a logarithmic error, whose size we will control explicitly. We prove the theorem in the following stages.

Here is the accounting plan. We will introduce three expected counts: A, the number of large queries; B, the number of pairs whose two labels have not appeared in a preceding small query; and W, the number of labels never appearing in a small query. Counting first appearances separately on each path will give

$$n - L \le B + W - A.$$

Thus we need upper bounds for A and for $B + W$. Comparing P with the truthful product law Q will control A. Comparing P with the artificial law R will control $B + W$, while retaining a useful subtraction for the large nodes. The separate entropy deficiencies will then combine within Γ. This paragraph describes the destination; each inequality is proved below.

9.1 Separate the entropy contributions

Recall $\Gamma = \sum_v P(v)(1 - h(\pi_v))$, with π_v the actual P-conditional negative probability. In both transcript comparisons, the first law in the divergence is P_Y. The tree-divergence formula therefore uses the contributions

$$P(v)\,\mathrm{kl}(\pi_v \| \rho_v^Q) \quad \text{and} \quad P(v)\,\mathrm{kl}(\pi_v \| \rho_v^R),$$

respectively. The reach weight is always $P(v)$, not $Q(v)$ or $R(v)$. Only the second binary parameter changes with the comparison experiment. Define

$$A = \sum_{v \text{ large}} P(v), \qquad \Gamma_{\mathrm{lg}} = \sum_{v \text{ large}} P(v)(1 - h(\pi_v)).$$

The first is the expected number of large queries, while the second is their contribution to entropy deficiency. Both use actual-law weights. We will keep these contributions separate from those of a second query class, rather than giving every class the full sum Γ.

9.2 Large pools under the truthful product law

Every P-reachable node is Q-reachable, since some valid input reaches it and every input subset has positive Q_X-mass. Fix such a node v, and work under the truthful experiment Q. Reaching it imposes two kinds of conditions. First, every coordinate in Z_v, the union of earlier negative pools, must be zero. Conditional on these fixed zeros, the remaining coordinates still have their independent Bernoulli-p law. Second, each earlier positive answer requires its pool to contain a defective. After deleting coordinates in Z_v, the intersection I_v of those positive requirements is an increasing event on the remaining coordinates.

Write $F_v = \{X_i = 0 \text{ for every } i \in Z_v\}$, and let $J_v = \{X \cap \widehat{S}_v = \varnothing\}$. The latter event is decreasing and uses only coordinates outside Z_v. Reaching v is exactly the event $F_v \cap I_v$; it has positive Q-probability. Apply the association inequality under the product law conditional on F_v:

$$\begin{aligned} \rho_v^Q &:= Q(\text{answer } 0 \mid \text{reach } v) = Q(J_v \mid F_v, I_v) \\ &\le Q(J_v \mid F_v) = q^{|\widehat{S}_v|}. \end{aligned} \tag{9.3}$$

Overlap among the positive pools, or between them and the current pool, is allowed. The remaining coordinates need not be independent after conditioning on I_v; association is what justifies the inequality without that additional independence.

At a large node,

$$\rho_v^Q \le (751/1000)^3 < \frac{17}{40}, \qquad \frac{1}{2} - \rho_v^Q > \frac{3}{40}.$$

The cubic inequality is certified by $751^3 = 423564751 < 425000000$. Also $\ln 2 < 7/10$, because

$$e^{7/10} > 1 + \frac{7}{10} + \frac{49}{200} + \frac{343}{6000} = \frac{12013}{6000} > 2.$$

The strict lower bound on the exponential follows, for example, by successive integration of $e^x > 1$ for $x > 0$.

Use the weighted inequality with $\omega = 1/10$. Its right side is greater than

$$\frac{2}{11}\frac{(3/40)^2}{7/10} = \frac{9}{6160} > \frac{1}{685}, \qquad 9 \cdot 685 = 6165 > 6160.$$

At every large node with $P(v) > 0$, multiplication by $P(v)$ gives

$$\frac{P(v)}{685} < \frac{P(v)}{10}(1 - h(\pi_v)) + P(v)\,\mathrm{kl}(\pi_v \| \rho_v^Q).$$

Sum over large nodes. The first terms total $\Gamma_{\mathrm{lg}}/10$; the second terms form part of the nonnegative tree-divergence sum $\mathrm{KL}(P_Y \| Q_Y) \le K$. Nodes with $P(v) = 0$ add zero. Weakening the resulting strict bound if necessary, we obtain

$$\frac{A}{685} \le \frac{\Gamma_{\mathrm{lg}}}{10} + K, \qquad \boxed{A \le \frac{137}{2}\Gamma_{\mathrm{lg}} + 685K.} \tag{9.4}$$

A large expected number of large queries thus requires entropy deficiency on those nodes or comparison cost. The latter is limited by the single whole-tree quantity K.

9.3 Fresh pairs under the reference law

A *fresh pair* is a normalized query with two labels neither of which has appeared in a preceding small normalized pool. They may have appeared in large pools. Define

$$B = \sum_{v \text{ fresh}} P(v), \qquad \Gamma_{\mathrm{fr}} = \sum_{v \text{ fresh}} P(v)(1 - h(\pi_v)).$$

Thus B is an expected number of queries and Γ_{fr} their entropy contribution. Fresh pairs are small, so their nodes are disjoint from large nodes and

$$\Gamma_{\mathrm{lg}} + \Gamma_{\mathrm{fr}} \le \Gamma. \tag{9.5}$$

Every P-reachable history is also R-reachable. To see this, choose a valid input reaching it truthfully. This input has positive Q_X-mass, it satisfies every small-answer constraint, and the finite prescribed sequence of fair large answers has positive probability. Formula (8.3) therefore gives positive reference mass. The same argument at terminal transcripts proves that P_Y is absolutely continuous with respect to R_Y. It also shows that every pair of positive $P_{X,Y}$-mass has positive $R_{X,Y}$-mass. All subsequent reference conditional probabilities and divergences are consequently well defined.

The reference posterior (8.4) leaves the two fresh coordinates independent Bernoulli-p. Consequently

$$\rho_v^R = R(\text{answer } 0 \mid \text{reach } v) = q^2 \ge (749/1000)^2 > \frac{14}{25}, \qquad \rho_v^R - \frac{1}{2} > \frac{3}{50}.$$

The exact square comparison is $749^2 = 561001 > 560000$. Earlier large positive answers in the truthful experiment might have constrained these labels. In R, those answers impose no input constraint on a fixed history; this is why the reference process is useful here.

9.4 The joint divergence has an exact value

For a valid input D, let $y(D)$ be its truthful terminal transcript, $\ell(D)$ its terminal leaf, and $a(D)$ the number of large queries along that path. The actual joint law gives $(D, y(D))$ mass $1/\binom{n}{d} = 2^{-H}$. In the reference experiment every small answer on this truthful path agrees with D, and every prescribed large answer has probability $1/2$. Thus

$$R_{X,Y}(D, y(D)) = Q_X(D)\, 2^{-a(D)} = 2^{-nh(p)-a(D)}. \tag{9.6}$$

The logarithmic ratio of actual to reference joint mass there is

$$\log \frac{2^{-H}}{2^{-nh(p)-a(D)}} = K + a(D).$$

Average under P, for which $\mathbb{E}_P a(X) = A$, to obtain

$$\boxed{\mathrm{KL}(P_{X,Y} \| R_{X,Y}) = K + A.} \tag{9.7}$$

This is a statement about input and transcript together. It is not yet a value for transcript divergence alone.

9.5 Labels untouched by small pools

On the truthful path of D, let $V(D)$ be the union of all small normalized pools. Define

$$w_1(D) = |D \setminus V(D)|, \quad w_0(D) = |([n] \setminus D) \setminus V(D)|, \quad w(D) = w_1(D) + w_0(D).$$

These count defective, nondefective, and total labels never seen in a small pool. They are quantities on one path, not yet averages. They may include labels used in large pools. Put $W = \mathbb{E}_P w(X)$.

Consider the following particular class of reference inputs:

$$\mathcal{A}_D = \{D' \subseteq [n] : D' \cap V(D) = D \cap V(D)\}.$$

Each agrees coordinate by coordinate with D on every small pool of the fixed transcript $y(D)$, so supplies all the prescribed small answers. The prescribed large answers depend only on fair reference bits and together have probability $2^{-a(D)}$. Thus the total reference mass reaching $y(D)$ from this class is

$$2^{-a(D)} Q_X(\mathcal{A}_D) = 2^{-a(D)} p^{d-w_1(D)} q^{n-d-w_0(D)}.$$

Here we fixed $d - w_1(D)$ defective and $n - d - w_0(D)$ nondefective coordinates inside $V(D)$. Each outside coordinate is unrestricted: summing its two possibilities contributes $p + q = 1$. There may be additional reference inputs producing the prescribed small OR answers without agreeing coordinate by coordinate on $V(D)$. Ignoring those inputs gives a lower bound:

$$\begin{aligned} R_Y(y(D)) &\geq 2^{-a(D)} p^{d-w_1(D)} q^{n-d-w_0(D)} \\ &= 2^{-a(D)} Q_X(D)\, p^{-w_1(D)} q^{-w_0(D)}. \end{aligned} \tag{9.8}$$

Inputs in $\mathcal{A}_D$ can have different total defective counts. This is permitted in the reference experiment; the real algorithm's exact-count promise has not been enlarged.

9.6 One bound shared by transcript and conditional divergence

Write

$$\mathcal{D}_R = \mathrm{KL}(P_Y \| R_Y), \qquad J = \mathbb{E}_P\left[w_1(X)\log\frac{1}{p} + w_0(X)\log\frac{1}{q}\right].$$

Here $\mathcal{D}_R$ concerns transcripts alone. The quantity J is an actual-law average of information associated with untouched labels. The joint chain rule is the exact identity

$$K + A = \mathcal{D}_R + \sum_{y:P_Y(y)>0} P_Y(y)\,\mathrm{KL}(P_{X|Y=y}\|R_{X|Y=y}). \tag{9.9}$$

At $y(D)$, correctness leaves exactly one valid input, D. Thus the actual conditional input law is the point mass at D. The reference conditional probability is

$$R(X = D \mid Y = y(D)) = \frac{R_{X,Y}(D, y(D))}{R_Y(y(D))} \le p^{w_1(D)} q^{w_0(D)}.$$

The inequality follows from (9.6) and (9.8): the numerator is known exactly and the denominator has a lower bound, producing an upper bound on the quotient. By the point-mass formula (5.1), the conditional divergence is at least

$$w_1(D)\log\frac{1}{p} + w_0(D)\log\frac{1}{q}.$$

Substitute its average into (9.9):

$$0 \le \mathcal{D}_R, \qquad \boxed{\mathcal{D}_R + J \le K + A.} \tag{9.10}$$

The direction has a concrete meaning. The full joint divergence is $K+A$. At least J of it remains in the conditional input comparison after the transcript is observed. Hence at most $K + A - J$ is available for transcript divergence. The lower bound on the conditional term produces an upper bound on the marginal term, and this will let us bound two counts together.

The direction can also be checked directly from leaf probabilities:

$$\log\frac{P_Y(y(D))}{R_Y(y(D))} \le K + a(D) - w_1(D)\log\frac{1}{p} - w_0(D)\log\frac{1}{q}.$$

Its P-average is exactly $\mathcal{D}_R \le K + A - J$.

We need a simple lower bound on the information per untouched label. In our interval $p \le q \le 751/1000 < 151/200$. The exact inequalities

$$(151/200)^2 < 4/7, \qquad (151/200)^5 < (4/7)^2(151/200) = 302/1225 < 1/4$$

show that $p^5, q^5 < 2^{-2}$. For the first rational comparison, $7 \cdot 22801 = 159607 < 160000$; for the last, $4 \cdot 302 = 1208 < 1225$. Taking logarithms gives $\log(1/p) > 2/5$ and $\log(1/q) > 2/5$. Therefore

$$J \ge \frac{2}{5}W. \tag{9.11}$$

9.7 Retain the contribution of fair artificial answers

At a large node the conditional reference answer is fair. Its contribution to the transcript divergence is $\mathrm{kl}(\pi_v\|1/2) = 1 - h(\pi_v)$. The total contribution of large nodes to $\mathcal{D}_R$ is therefore exactly Γ_{lg}. All other node divergences are nonnegative. In particular the sum over fresh-pair nodes is at most $\mathcal{D}_R - \Gamma_{\mathrm{lg}}$.

At a fresh pair use the weighted inequality with $\omega = 64$. Its right side is greater than

$$\frac{128}{65}\frac{(3/50)^2}{7/10} = \frac{576}{56875} > \frac{1}{99}, \qquad 576 \cdot 99 = 57024 > 56875.$$

At a fresh-pair node with $P(v) > 0$, we consequently have

$$\frac{P(v)}{99} < 64P(v)(1 - h(\pi_v)) + P(v)\,\mathrm{kl}(\pi_v\|\rho_v^R).$$

Sum with the actual-law reach weights. The first terms total $64\Gamma_{\mathrm{fr}}$. The divergence terms from all large nodes total exactly Γ_{lg}, and every other node contributes nonnegatively. The fresh-pair divergence sum is therefore at most $\mathcal{D}_R - \Gamma_{\mathrm{lg}}$, yielding

$$\frac{B}{99} \le 64\Gamma_{\mathrm{fr}} + \mathcal{D}_R - \Gamma_{\mathrm{lg}}. \tag{9.12}$$

Combine this with (9.10), keeping its conditional term:

$$\frac{B}{99} + J \le 64\Gamma_{\mathrm{fr}} + K + A - \Gamma_{\mathrm{lg}}.$$

Multiply by 99. Since $99J \ge (198/5)W \ge W$, we get

$$\boxed{B + W \le 6336\Gamma_{\mathrm{fr}} + 99K + 99A - 99\Gamma_{\mathrm{lg}}.} \tag{9.13}$$

The estimate controls fresh pairs and untouched labels jointly. The subtraction of Γ_{lg} is useful: the large nodes have already used that exact portion of the transcript divergence, leaving only the remainder for fresh pairs.

9.8 Count first appearances on a single path

Fix a valid D. Let $t(D)$ be its number of tests and $b(D)$ its number of fresh pairs. Its number of small queries is $t(D) - a(D)$. Count a label when it first appears in a small pool:

- an empty pool adds none;
- a singleton adds at most one;
- a pair with a previously seen label adds at most one;
- a fresh pair adds two, one more than that general allowance.

Consequently

$$|V(D)| \le t(D) - a(D) + b(D).$$

This is a pathwise combinatorial inequality, valid with arbitrary overlaps. Since $|V(D)| = n - w(D)$, average under P to obtain

$$n - W \le L - A + B, \qquad n - L \le B + W - A. \tag{9.14}$$

The probability law is the same for all the expectations on this line.

9.9 Finish the coefficient calculation

Insert (9.13) into (9.14), and then use (9.4):

$$\begin{aligned} n - L &\le 6336\Gamma_{\mathrm{fr}} + 99K + 99A - 99\Gamma_{\mathrm{lg}} - A \\ &= 6336\Gamma_{\mathrm{fr}} + 99K + 98A - 99\Gamma_{\mathrm{lg}} \\ &\le 6336\Gamma_{\mathrm{fr}} + \left(98 \cdot \frac{137}{2} - 99\right)\Gamma_{\mathrm{lg}} + (99 + 98 \cdot 685)K \\ &= 6336\Gamma_{\mathrm{fr}} + 6614\Gamma_{\mathrm{lg}} + 67229K. \end{aligned}$$

The arithmetic is $98 \cdot 137/2 = 6713$ and $98 \cdot 685 = 67130$. Because $6336 \le 6614$ and $\Gamma_{\mathrm{fr}} + \Gamma_{\mathrm{lg}} \le \Gamma$,

$$n - L \le 6614\Gamma + 67229K.$$

Now $L = H + \Gamma$, so $n - H = (n - L) + \Gamma$ gives

$$n - H \le 6615\Gamma + 67229K,$$

which is (9.2). An optimal worst-case tree has $L \le M(d, n)$. Replacing $L - H$ by the larger $M(d, n) - H$ preserves the lower-bound inequality and proves the last assertion of Theorem 9.1. □

All query types have been included: large pools through the product comparison, fresh pairs through the reference posterior, and other small pools through nonnegative omitted terms and the first-appearance count. The finite-tree identities allow one-sided feasible answers, empty pools, and unequal leaf depths. The artificial process has served only as a comparison law.

10 A structural relation between different testing problems

The information bound applies near density 1/4. To use it for Theorem 1.4, we need a relation that moves between complete families of different sizes. This section is entirely deterministic. Its simulations use actual pooled-OR queries; no independent input law or artificial answers are involved.

10.1 Complexity of a candidate family

A *candidate family* $\mathcal{F}$ is a nonempty finite set of possible defective subsets of a specified ground set. The algorithm is told the family, and must identify its member using legal OR tests. A legal query is still any subset S of that ground set, with answer one on candidate D exactly when $S \cap D \neq \varnothing$. Knowing the family does not mean knowing which member is actual, and does not permit arbitrary binary questions about its members. Let $\mathrm{comp}(\mathcal{F})$ denote the minimum worst-case depth. Thus

$$M(d, n) = \mathrm{comp}\left(\binom{[n]}{d}\right).$$

This notation lets us discuss the families left after tests, which need not be complete families of equal-size subsets.

Lemma 10.1 (Elementary reductions)**.** *The following operations have the stated effects:*

1. *Restriction to a nonempty subfamily cannot increase complexity.*

2. *Relabeling items preserves complexity.*

3. *Removing labels known to be nondefective preserves complexity.*

4. *Adjoining the same fixed defective labels to every candidate preserves complexity.*

5. *A tree may be reduced so that both children of every internal node have nonempty candidate families, without increasing its depth.*

Proof. A tree correct for a family is also correct for each subfamily; impossible paths may be removed. This proves the first assertion and fixes its direction. For relabeling, apply the same bijection to every candidate and query. All intersection answers are preserved, and the inverse bijection gives the reverse comparison.

A known nondefective label can be deleted from any pool without changing an answer. Conversely an algorithm that ignores that label still works when it is present. If every candidate contains a fixed defective label, any query containing it is constantly positive. Suppress such a query, following its positive branch without a test. Other queries ask only about the remaining unknown part. In the other direction, a tree for that unknown part also identifies the family with the fixed labels adjoined. This proves equality of complexities.

Finally, if one child of an internal node has no possible candidates, the answer is forced on every remaining input. Suppress that query and continue into the other child. Repeating this operation terminates in a finite tree. Leaves remain correct and path lengths do not increase. □

These reductions use information already contained in the specified candidate family. They do not give the algorithm extra knowledge about which member is the actual input. In a reduced tree an internal node cannot have only one surviving candidate, since then no query could have two nonempty children.

10.2 The endpoint problems

Lemma 10.2.

$$M(N-1,N) = N-1 \quad (N \geq 2), \qquad M(1,2) = 1, \qquad M(1,3) = 2.$$

Proof. When exactly one item is good, every pool of size at least two is positive. Empty queries are constant too. A singleton query can ask whether a particular label is the one good item. For the lower bound, follow a feasible adversarial branch. While at least two possible locations of the nondefective item remain, answer a singleton test positively whenever its label is still a candidate location. This declares that label defective, eliminates exactly one possible location, and leaves at least one candidate. Pools of size at least two, and singletons already eliminated as possible locations, are forced positive and eliminate nothing further. An empty pool is forced negative and likewise eliminates nothing. Starting with N possible locations, this feasible branch therefore requires at least $N-1$ tests before only one location remains. Individual testing achieves that number.

With one defective among two or three items, the numbers of candidates are respectively two and three. A binary tree needs at least one or two tests by leaf counting. Individual testing achieves those bounds. □

10.3 Two distinguished labels

The next identity is the key structural step. It explains the complexity of a family in which we know that at least one of two specified labels is defective.

Let U contain m ordinary labels, and let ξ, η be two additional distinct labels. Fix $1 \le k < m$. Define

$$\mathcal{F} = \binom{U}{k}, \qquad \mathcal{G} = \binom{U}{k-1}.$$

For this subsection only, ξD means $\{\xi\} \cup D$, and $\xi\eta E$ means $\{\xi, \eta\} \cup E$. Consider

$$\mathcal{J} = \{\xi D : D \in \mathcal{F}\} \cup \{\eta D : D \in \mathcal{F}\} \cup \{\xi\eta E : E \in \mathcal{G}\},$$
$$\mathcal{C} = \mathcal{F} \cup \{\eta E : E \in \mathcal{G}\} = \binom{U \cup \{\eta\}}{k}.$$

Every member of $\mathcal{J}$ has exactly $k+1$ defectives and meets $\{\xi, \eta\}$. Its three types are:

Distinguished defectives	Ordinary defectives	Candidate form
Only ξ	k labels from U	ξD
Only η	k labels from U	ηD
Both	$k-1$ labels from U	$\xi\eta E$

The family $\mathcal{C}$ is an ordinary complete problem: k defectives among $m+1$ labels.

Lemma 10.3 (Two-label identity)**.**

$$\mathrm{comp}(\mathcal{J}) = 1 + \mathrm{comp}(\mathcal{C}) = 1 + M(k, m+1).$$

The easy direction. Test $\{\xi\}$. On a positive answer the remaining family is $\{\xi C : C \in \mathcal{C}\}$, whose complexity is $\mathrm{comp}(\mathcal{C})$ after deleting the fixed defective ξ. On a negative answer the family is $\{\eta D : D \in \mathcal{F}\}$, with complexity $\mathrm{comp}(\mathcal{F}) \le \mathrm{comp}(\mathcal{C})$ by restriction. Thus $\mathrm{comp}(\mathcal{J}) \le 1 + \mathrm{comp}(\mathcal{C})$.

The reverse direction must work even if a proposed tree never begins by testing a distinguished label. We will convert any reduced tree for $\mathcal{J}$, of height s, into a legal tree for $\mathcal{C}$ of height at most $s-1$.

What can ordinary queries distinguish? A query contained in U will be called ordinary. After a prefix made only of ordinary queries, let $\mathcal{A} \subseteq \mathcal{F}$ and $\mathcal{B} \subseteq \mathcal{G}$ be the ordinary subsets consistent with those answers. The surviving family in $\mathcal{J}$ is

$$\{\xi D : D \in \mathcal{A}\} \cup \{\eta D : D \in \mathcal{A}\} \cup \{\xi\eta E : E \in \mathcal{B}\}. \tag{10.1}$$

In particular, ξD and ηD cannot be distinguished by ordinary queries: both meet an ordinary pool exactly when D does.

There is also a useful relation between the two subset sizes. If E, E' are distinct members of $\mathcal{B}$, choose $z \in E' \setminus E$. Then $E \cup \{z\}$ belongs to $\mathcal{A}$. To verify this, every earlier positive pool meets E, and therefore also meets $E \cup \{z\}$. Every earlier negative pool avoids E', hence avoids z; it also avoids E. So all negative answers are preserved. The enlarged set has size k, as required. We have proved

$$\mathcal{A} = \varnothing, \quad \mathcal{B} \neq \varnothing \quad \Longrightarrow \quad |\mathcal{B}| = 1. \tag{10.2}$$

A deepest ordinary-only path is impossible. Suppose a reduced $\mathcal{J}$-tree of height s has a leaf at depth s identifying $\xi\eta E$, and all queries on its path are ordinary. Consider its parent, at depth $s-1$. Its two nonempty children must both be leaves, so at most two candidates can survive at the parent. If its $\mathcal{A}$ is nonempty, any $D \in \mathcal{A}$ supplies the distinct candidates $\xi D, \eta D, \xi\eta E$, giving at least three. If $\mathcal{A}$ is empty, (10.2) makes the surviving family a singleton, contrary to an internal node of a reduced tree. Both possibilities are impossible.

Therefore an ordinary-only leaf of the form $\xi\eta E$ has depth at most $s-1$. This observation will handle paths where there is no special query to skip.

The simulation when a distinguished query occurs. The idea is to assign each possible input C of the new problem an injective representative $\phi(C)$ in the old problem. Once this assignment is selected, the simulation maintains the following invariant: every currently possible C has its representative among the surviving candidates of the old tree, and every query actually asked has the same answer on C as the old query has on $\phi(C)$. An old query may be skipped only when its answer is known for every current representative. The new transcript then follows a well-defined path through the old tree, without knowing which input is actual. We will choose one of two assignments at the first distinguished query; both preserve every earlier ordinary answer.

Run a new algorithm on an unknown $C \in \mathcal{C}$. Until the first query containing a distinguished label, use the same ordinary queries as the $\mathcal{J}$-tree. The possible inputs in $\mathcal{C}$ at that point are $\mathcal{A} \cup \{\eta E : E \in \mathcal{B}\}$; the corresponding candidates in (10.1) have the same ordinary answers.

A query containing both ξ and η is constantly positive on $\mathcal{J}$, so it is absent in the reduced tree. The first nonordinary query therefore contains exactly one of them. There are two cases.

1. If it contains ξ, enter its positive child without testing. From that point represent the possible inputs by
$$D \in \mathcal{F} \longmapsto \xi D, \qquad \eta E \in \mathcal{C} \longmapsto \xi\eta E.$$
Every representative satisfies the previous ordinary answers and contains ξ, so it satisfies the skipped positive answer. Later queries containing ξ are skipped into their positive children. A query avoiding ξ is asked unchanged on $U \cup \{\eta\}$. Its OR answer equals the representative's answer, because the only added defective label is ξ.

2. If the first query contains η, instead use
$$D \in \mathcal{F} \longmapsto \eta D, \qquad \eta E \in \mathcal{C} \longmapsto \xi\eta E.$$
Again skip that query into its positive child, and skip any later query containing η. For a later query avoiding η, keep its ordinary labels and replace ξ, if present, by the label η of the $\mathcal{C}$-problem. To check this replacement, write the old query as S or $S \cup \{\xi\}$, where $S \subseteq U$. For a new input $D \in \mathcal{F}$, its representative is ηD: η is absent from the new input and ξ is absent from the representative. For a new input ηE, its representative is $\xi\eta E$: both of these compared labels are present. In either case the ordinary part S meets precisely the same ordinary defectives. Replacing ξ by η therefore preserves the OR answer, including when S is empty.

These representatives are part of a mathematical simulation, not information revealed about the actual input. The choices of which case applies depend only on the query reached in the original tree. All asked questions are subsets of $U \cup \{\eta\}$, and are ordinary pooled-OR tests. The representation

in either case is injective, so the identifying output of the old tree uniquely determines the input of the new one. The original family has the known count $k+1$, the simulated family the known count k; each specified representation adds exactly one defective.

Check every path length. Any simulated path that reaches a distinguished query skips at least that first query, so has length at most $s-1$. If no such query occurs, it ends at an ordinary-only leaf. That leaf cannot identify ξD or ηD, which are indistinguishable by ordinary queries. It identifies $\xi\eta E$, and its depth is at most $s-1$ by the parent argument above. Its corresponding $\mathcal{C}$-input is uniquely ηE. Thus every simulated path has length at most $s-1$. We have $\mathrm{comp}(\mathcal{C}) \le s-1$, completing the lower bound and the proof of Lemma 10.3. □

When $k=1$, the family $\mathcal{G}$ consists of the empty set. The same proof works: its extension argument requires no pair of distinct members in that case, and a nonempty $\mathcal{B}$ already has just one member.

10.4 The recurrence and its direction

Proposition 10.4. *For integers $r \ge 2$ and $N \ge r+2$,*

$$\boxed{M(r,N) \ge \min\{1+M(r,N-1),\ 2+M(r-1,N-1)\}.} \tag{10.3}$$

Proof. Take a reduced optimal tree for the complete family of r-subsets of an N-element set. Its root cannot be a constant query.

If its pool is a singleton, the negative child is a complete $(r, N-1)$ problem. The root plus that child gives height at least $1+M(r,N-1)$. The positive child may be harder, but ignoring it cannot invalidate this lower bound on the maximum of child depths.

If its pool has at least two labels, choose distinct ξ, η inside it. Every r-subset meeting this pair gives a positive root answer, irrespective of the other labels in the pool. Thus the positive child contains the family $\mathcal{J}$ of Lemma 10.3, with U the other $N-2$ labels and $k = r-1$. The conditions $r \ge 2, N \ge r+2$ ensure $1 \le k < |U|$. By restriction and the identity, the child's complexity is at least $1+M(r-1,N-1)$. Add the root to get $2+M(r-1,N-1)$.

Every nonconstant root belongs to one of these cases. Each tree is at least as deep as the bound for its case, and hence at least as deep as the smaller of the two bounds. This explains the minimum in (10.3). □

The argument did not restrict the root to singleton or pair pools: the second case includes every pool with two or more labels. Nor did it assume a special structure for later queries. Both possible outcomes remain part of a legal tree, even though one child can suffice to certify a lower bound.

11 Turning a supposed saving into a contradiction

We now combine the two parts of the proof. The recurrence will carry a supposed improvement over individual testing to a smaller complete-family problem near density $1/4$. The information inequality will say that this smaller problem cannot have the amount of saving that the recurrence requires.

11.1 Measure the saving

For $0 < r < N$, define

$$\Delta(r, N) = N - 1 - M(r, N).$$

This is a nonnegative integer by individual testing. It is the number of tests the optimal worst-case bound saves compared with $N - 1$. A counterexample to the desired equality has $\Delta(r, N) \geq 1$.

Rearrange the recurrence carefully. The two expressions on its right are

$$1 + M(r, N - 1) = N - 1 - \Delta(r, N - 1)$$

and

$$2 + M(r - 1, N - 1) = N - \Delta(r - 1, N - 1).$$

Subtracting their minimum from $N - 1$ gives

$$\Delta(r, N) \leq \max\{\Delta(r, N - 1),\ \Delta(r - 1, N - 1) - 1\} \tag{11.1}$$

when $r \geq 2, N \geq r + 2$.

Suppose the current saving is at least s_0. The maximum on the right must be at least s_0. Therefore at least one of the following is true:

Move to	Change of parameters	Required saving there
$(r, N - 1)$	Remove one from the size only	At least s_0
$(r - 1, N - 1)$	Remove one from rank and size	At least $s_0 + 1$

Here rank means the defective count. A move lowering it increases the forced saving by one.

This sequence of parameter pairs is a mathematical consequence of lower bounds. It is not an actual test execution removing known items for free. We are not claiming to know which items to delete from an unknown input. The recurrence asserts that one of two separate, completely specified testing problems must have the indicated complexity.

11.2 Start from an alleged improvement

Assume, for a contradiction, that

$$n \geq 2^{31}, \qquad d/n \geq c_\star, \qquad \Delta(d, n) \geq 1.$$

Use (11.1) to choose an available move of the indicated kind at each stage. After j moves lowering rank and size, and k_{size} moves lowering size only, the current pair satisfies

$$r = d - j, \qquad N = n - j - k_{\text{size}}, \qquad \Delta(r, N) \geq 1 + j. \tag{11.2}$$

This is proved by induction on moves: size-only moves preserve the saving lower bound; rank-lowering moves increase it by one.

Stop at the first pair for which $p = r/N \leq 1/4$. The starting density is above $1/4$, since $c_\star > 1/3 > 1/4$; the first inequality follows from $3 \cdot 369056 > 1000000$.

11.3 Why the sequence must stop at the desired density

The recurrence is unavailable only if $r = 1$ or $N = r + 1$. While the density is greater than $1/4$, the case $r = 1$ means $N = 2$ or 3. The other unavailable case is the one-good-item problem $N = r + 1$. All these states have $\Delta = 0$ by Lemma 10.2, contradicting the positive lower bound in (11.2). Thus an unavailable state cannot end our sequence above $1/4$.

Each move decreases N by one, so the sequence cannot continue indefinitely. Every permitted successor is admissible: the recurrence guards $r \geq 2, N \geq r + 2$ ensure positive new rank and new rank smaller than new size. It follows that the specified stopping pair exists.

A size-only move raises the ratio r/N. Therefore the first crossing below $1/4$ must be a rank-lowering move. Its predecessor was $(r + 1, N + 1)$, with density strictly above $1/4$. From

$$\frac{r+1}{N+1} > \frac{1}{4} \quad \Longrightarrow \quad 4r > N - 3$$

we obtain the useful rounding control

$$\frac{1}{4} - \frac{3}{4N} < p = \frac{r}{N} \leq \frac{1}{4}. \tag{11.3}$$

There is no need to assume that N is divisible by four. Equality $p = 1/4$ is allowed at the stopping state.

11.4 The remaining instance is still large

Since $M(r, N) \geq 0$, we have $\Delta(r, N) \leq N - 1$. Combining this with (11.2) gives $j \leq N - 2$. At the stopping state $r \leq N/4$, so

$$d = r + j \leq \frac{N}{4} + N - 2 = \frac{5}{4}N - 2.$$

Consequently

$$N \geq \frac{4}{5}(d + 2) > \frac{n}{4} \geq 2^{29}. \tag{11.4}$$

For the strict middle inequality, use $d \geq c_\star n$ and $c_\star > 5/16$, since $16 \cdot 369056 = 5904896 > 5000000$. Thus $(4/5)c_\star > 1/4$.

Because $N > 2^{29} > 750$, the rounding term $3/(4N)$ is less than $1/1000$. Equation (11.3) therefore places the stopped density in

$$\frac{249}{1000} < p \leq \frac{1}{4}.$$

This is inside the interval of the whole-tree inequality. There are no small residual instances left to check.

11.5 How much saving is forced at the stopped state?

Let $c = d/n$ be the original density. This is an arbitrary number at least $c_\star$, not necessarily equal to it. Since $k_{\text{size}} \geq 0$,

$$N \leq n - j = n - d + r = n - d + pN.$$

Rearranging, with $c < 1$, gives

$$\frac{n}{N} \geq \frac{1 - p}{1 - c}.$$

Since $j = d - r$,

$$\frac{j}{N} = \frac{d}{N} - p \geq \frac{c(1-p)}{1-c} - p = \frac{c-p}{1-c}. \tag{11.5}$$

For fixed $p < 1$, the derivative of the last expression with respect to c is $(1-p)/(1-c)^2 > 0$. For fixed $c < 1$, its derivative with respect to p is $-1/(1-c) < 0$. Using $c \geq c_\star$ and $p \leq 1/4$, we conclude

$$\frac{j}{N} \geq \frac{c_\star - 1/4}{1 - c_\star} = \frac{119056}{630944} = \frac{7441}{39434}.$$

Define

$$g_\star = \frac{7441}{39434}. \tag{11.6}$$

The tracked saving is therefore at least

$$\Delta(r, N) \geq 1 + g_\star N. \tag{11.7}$$

This is why all starting densities above $c_\star$ are covered: increasing the original density only strengthens the lower bound on j/N.

11.6 What the information bound allows

Apply Theorem 9.1 to an optimal tree for the stopped (r, N) problem. Here use

$$H = \log \binom{N}{r}, \qquad K = Nh(p) - H.$$

The theorem and expected depth at most worst-case depth give

$$M(r, N) \geq H + \frac{N - H - 67229K}{6615}.$$

Subtract from $N - 1$, and show each coefficient:

$$\begin{aligned}
\Delta(r, N) &\leq N - 1 - H - \frac{N-H}{6615} + \frac{67229}{6615}K \\
&= \frac{6614}{6615}(N - H) + \frac{67229}{6615}K - 1 \\
&= \frac{6614}{6615}N(1 - h(p)) + \frac{6614 + 67229}{6615}K - 1.
\end{aligned}$$

Put

$$\alpha = \frac{6614}{6615}, \qquad \beta = \frac{73843}{6615} < 12.$$

The last comparison follows from $73843 < 12 \cdot 6615 = 79380$. Using $K \leq \log(N+1)$, we obtain

$$\Delta(r, N) \leq \alpha N(1 - h(p)) + \beta \log(N+1) - 1. \tag{11.8}$$

We next replace the nearby density p by exactly $1/4$, paying only a constant for the rounding. Define

$$e_{\text{quarter}} = 1 - h(1/4).$$

For $249/1000 \leq u \leq 1/4$,

$$h'(u) = \log \frac{1-u}{u} < 2,$$

since $(1-u)/u \le 751/249 < 4$. Integration from p to $1/4$, together with (11.3), yields

$$h(1/4) - h(p) \le 2(1/4 - p) < \frac{3}{2N}.$$

It also holds when $p = 1/4$, because then the left side is zero. Thus

$$N(1 - h(p)) < Ne_{\text{quarter}} + \frac{3}{2}. \tag{11.9}$$

11.7 The numerical margin and the contradiction

The exact arithmetic needed now is isolated in the next lemma. Its proof, including finite logarithm series and their remainder bounds, is given in Appendix A.

Lemma 11.1 (Two numerical certificates)**.** *With the constants above,*

$$g_\star - \alpha e_{\text{quarter}} > \frac{1}{1000000}. \tag{11.10}$$

For every real $N \ge 2^{29}$,

$$12\log(N+1) < \frac{N}{1000000}. \tag{11.11}$$

Combine the lower saving (11.7), the upper saving (11.8), and the rounded entropy estimate:

$$1 + g_\star N \le \Delta(r, N) < \alpha N e_{\text{quarter}} + \frac{3\alpha}{2} + \beta\log(N+1) - 1.$$

Move the linear terms to the left and the two constant ones to the right:

$$(g_\star - \alpha e_{\text{quarter}})N < \beta\log(N+1) + \frac{3\alpha}{2} - 2.$$

Since $0 < \alpha < 1$, the final constant is negative. Hence a counterexample would require

$$(g_\star - \alpha e_{\text{quarter}})N < \beta\log(N+1). \tag{11.12}$$

But Lemma 11.1, $\beta < 12$, and $N > 2^{29}$ give the incompatible chain

$$(g_\star - \alpha e_{\text{quarter}})N > \frac{N}{1000000} > 12\log(N+1) > \beta\log(N+1).$$

This contradiction rules out $\Delta(d, n) \ge 1$. As $\Delta(d, n)$ is a nonnegative integer, it must equal zero:

$$M(d, n) = n - 1.$$

Together with Proposition 1.3, this proves Theorem 1.4. □

11.8 What each constant does

The proof separates several roles that are easy to confuse.

- The intermediate density $1/4$ makes large pools and fresh pairs biased in useful directions under their respective comparison laws. The small interval around it absorbs the integer crossing error.

- The coefficient 6615 comes from the tree accounting. It produces $\alpha = 1 - 1/6615 < 1$ in the upper bound on savings.
- The starting threshold $c_\star$ determines $g_\star = (c_\star - 1/4)/(1 - c_\star)$, the saving forced per remaining item by the recurrence.
- The margin $g_\star - \alpha e_{\text{quarter}} > 10^{-6}$ means that the forced linear saving exceeds what the information estimate permits, apart from its error term.
- The exact-count comparison contributes $K \le \log(N+1)$, giving a logarithmic error. The cutoff $n \ge 2^{31}$ guarantees a residual $N > 2^{29}$, where the linear margin dominates that error.

The numerical certificate is used to establish these inequalities, not to guess a pattern from small instances.

12 What has, and has not, been proved

The conclusion is an exact worst-case equality, for every admissible integer pair in the stated range. The proof allowed arbitrary legal pools, arbitrary overlap, and full adaptivity. The average used in the probability argument was always over valid exact-count inputs. The independent and artificial laws were comparison measures, with every change of law accounted for by an identity or inequality.

The expectation estimate alone was not the final result. It supplied an upper bound on possible savings at a smaller instance. The structural recurrence forced an incompatible lower bound on those savings from any proposed original counterexample. This is the step that upgrades the information estimate to the exact answer $n - 1$.

There is no assertion that the same threshold theorem holds for all $n < 2^{31}$. The elementary endpoint results proved earlier do hold at their stated smaller sizes, but they are not a substitute for a general small-population theorem. The constants in the argument are sufficient constants; their optimality is not claimed.

The conjectural boundary is density $1/3$, often called the Hu–Hwang–Wang conjecture [2]. Since $3 \cdot 369056 > 1000000$, the theorem explained here stops strictly above that boundary. No resolution at $1/3$ follows from this proof.

A Exact certificates for the numerical step

This appendix proves Lemma 11.1. Its purpose is narrow: to certify a positive linear margin and the population size at which that margin dominates the logarithmic correction. Decimal approximations are not used as evidence.

A.1 A logarithm series with a proved remainder

For $0 < t < 1$, the derivative of $\ln((1+t)/(1-t))$ is $2/(1-t^2)$. The finite geometric identity gives

$$\frac{1}{1-u^2} = \sum_{j=0}^{s} u^{2j} + \frac{u^{2s+2}}{1-u^2}.$$

Integrate from zero to t, multiply by two, and use that the logarithm is zero at zero:

$$\ln\frac{1+t}{1-t} = 2\sum_{j=0}^{s}\frac{t^{2j+1}}{2j+1} + R_s(t). \tag{A.1}$$

The remainder is the integral of $2u^{2s+2}/(1-u^2)$. It is positive, and $1-u^2 > 1-t^2$ for $0 \le u < t$. Therefore

$$0 < R_s(t) < \frac{2t^{2s+3}}{(2s+3)(1-t^2)}. \tag{A.2}$$

At $t = 1/3$ the logarithm in (A.1) is $\ln 2$; at $t = 1/5$ it is $\ln(3/2)$.

A.2 A finite rational table

Set $T_0 = 10^{10}$. Every row in the following table has positive integers u, f satisfying

$$uf \le 2T_0 < u(f+1).$$

Equivalently, $f/T_0 \le 2/u < (f+1)/T_0$. Thus the table gives lower and upper rational bounds for each series term; it is a certificate consisting only of integer multiplications and comparisons.

	For $\ln 2$, $t = 1/3$			For $\ln(3/2)$, $t = 1/5$	
j	$u = (2j+1)3^{2j+1}$	f	j	$u = (2j+1)5^{2j+1}$	f
0	3	6666666666	0	5	4000000000
1	81	246913580	1	375	53333333
2	1215	16460905	2	15625	1280000
3	15309	1306421	3	546875	36571
4	177147	112900	4	17578125	1137
5	1948617	10263	5	537109375	37
6	20726199	964			
7	215233605	92			
8	2195382771	9			
	Sum of f:	6931471800		Sum of f:	4054651078

For example, in the row $u = 81$ the product is $81 \cdot 246913580 = 19999999980$, while adding another 81 exceeds $20000000000 = 2T_0$. The displayed inequalities for the other rows have exactly the same form.

The remainder bounds (A.2) give

$$R_8(1/3) < \frac{1}{76 \cdot 3^{17}} = \frac{1}{9814652388} < \frac{2}{T_0},$$

and

$$R_5(1/5) < \frac{1}{156 \cdot 5^{11}} = \frac{1}{7617187500} < \frac{2}{T_0}.$$

In the first sum the error from replacing the nine terms by their table bounds is less than $9/T_0$. In the second it is less than $6/T_0$. Hence

$$\frac{6931471800}{T_0} < \ln 2 < \frac{6931471800 + 9 + 2}{T_0} < \frac{6931471900}{T_0},$$

and

$$\frac{4054651000}{T_0} < \frac{4054651078}{T_0} < \ln(3/2) < \frac{4054651078+6+2}{T_0} < \frac{4054651100}{T_0}.$$

In particular,

$$\frac{69314718}{10^8} < \ln 2 < \frac{69314719}{10^8}, \qquad \frac{40546510}{10^8} < \ln(3/2) < \frac{40546511}{10^8}. \tag{A.3}$$

These also certify the simpler bound $\ln 2 > 1/2$.

A.3 Bound the entropy gap at one quarter

Let $\theta = \log_2 3$. Since $\theta = 1 + \ln(3/2)/\ln 2$, the upper bound on the numerator and the lower bound on the denominator in (A.3) yield

$$\theta < 1 + \frac{40546511}{69314718} < \frac{1584963}{1000000}.$$

The last comparison has the exact integer certificate

$$584963 \cdot 69314718 - 1000000 \cdot 40546511 = 34385434 > 0.$$

From $h(1/4) = 2 - 3\theta/4$, we obtain

$$e_{\text{quarter}} = 1 - h(1/4) = \frac{3}{4}\theta - 1 < \bar{e} := \frac{754889}{4000000}. \tag{A.4}$$

A.4 The positive margin

Recall $g_\star = 7441/39434$ and $\alpha = 6614/6615$. Exact subtraction gives

$$g_\star - \bar{e} = -\frac{4292826}{157736000000} > -\frac{55}{2000000}.$$

After multiplication by the positive denominator, the final inequality is $4292826 < 4337740$. Also

$$\frac{\bar{e}}{6615} = \frac{754889}{26460000000} > \frac{57}{2000000},$$

whose certificate is $754889 > 754110$. Now combine these as parts of one expression:

$$\begin{aligned} g_\star - \alpha e_{\text{quarter}} &> g_\star - \alpha\bar{e} \\ &= (g_\star - \bar{e}) + \frac{\bar{e}}{6615} \\ &> -\frac{55}{2000000} + \frac{57}{2000000} = \frac{1}{1000000}. \end{aligned}$$

The first inequality uses $\alpha > 0$ and $e_{\text{quarter}} < \bar{e}$. This proves (11.10).

A.5 The cutoff

For real $N \geq 2^{29}$, define

$$F(N) = \frac{N}{1000000} - 12\log(N+1).$$

At $N = 2^{29} = 536870912$, the first term exceeds 536. Since $2^{29} + 1 < 2^{30}$, the second term is less than $12 \cdot 30 = 360$. Thus $F(2^{29}) > 0$. Furthermore, using $\ln 2 > 1/2$,

$$F'(N) = \frac{1}{1000000} - \frac{12}{(N+1)\ln 2} > \frac{1}{1000000} - \frac{24}{N+1} > 0.$$

The last inequality holds because $N + 1 \geq 2^{29} + 1 > 24000000$. So $F(N) > 0$ for the entire range, proving (11.11). This completes the proof of Lemma 11.1.

B Notation guide

The following guide collects the main notation; definitions and justifications are in the body of the exposition.

Symbol	Meaning
n, d	Original population size and exact defective count.
D	A particular defective subset.
$X,\ X_i$	Random defective subset and its indicator at label i.
$S_v, \widehat{S}_v$	Original and normalized pools at node v.
Z_v	Union of preceding pools assigned negative answers on the node's fixed prefix.
$\tau,\ y,\ \ell$	Prefix transcript, terminal transcript, and terminal leaf.
Y, T	Random terminal transcript and number of tests in an execution.
$P_X, P_Y, P_{X,Y}$	Input, transcript, and joint laws for uniform exact-count inputs with truthful answers.
$Q_X, Q_Y, Q_{X,Y}$	Corresponding laws for independent inputs with truthful answers.
$R_X, R_Y, R_{X,Y}$	Corresponding laws for independent inputs with artificial fair answers at large normalized pools. Here $R_X = Q_X$.
$P(v)$	Probability under P of reaching node v.
$\pi_v, \rho_v^Q, \rho_v^R$	Conditional negative probabilities at v under P, Q, R, respectively.
$\mathsf{H},\ h,\ \mathrm{KL},\ \mathrm{kl}$	Finite-law entropy, binary entropy, relative entropy, and binary relative entropy. All use base-two logarithms.
p, q	The current exact defective fraction and its complement $q = 1 - p$.
H	$\log \binom{n}{d}$, entropy of the uniform valid-input law.
K	$nh(p) - H = \mathrm{KL}(P_X \Vert Q_X)$, the exact-count comparison cost.
L, Γ	Actual expected test count $\mathbb{E}_P T$ and its excess $L - H$.
$\Gamma_{\mathrm{lg}}, \Gamma_{\mathrm{fr}}$	Parts of Γ contributed by large queries and fresh pairs.
$a(D), b(D), t(D)$	Large-query count, fresh-pair count, and test count on the truthful path of D.
A, B	The P-expected large-query and fresh-pair counts.
$V(D), w_1(D), w_0(D)$	Union of small pools on the truthful path, and counts of defective and good labels outside that union.

Symbol	Meaning
W	$\mathbb{E}_P(w_1 + w_0)$, expected number of untouched labels.
J	$\mathbb{E}_P[w_1 \log(1/p) + w_0 \log(1/q)]$, a lower bound on averaged conditional input divergence.
$\mathcal{D}_R$	$\mathrm{KL}(P_Y \Vert R_Y)$, reference transcript divergence.
$\mathrm{comp}(\mathcal{F})$	Minimum worst-case depth for a specified nonempty candidate family $\mathcal{F}$.
ξ, η	The two distinguished labels in the structural identity.
r, N	Rank and population size at a smaller recurrence instance.
$\Delta(r, N)$	$N - 1 - M(r, N)$, saving compared with individual testing.
j, k_{size}	Counts of rank-and-size moves and size-only moves in the recurrence sequence.
$c_\star, g_\star$	369056/1000000 and $(c_\star - 1/4)/(1 - c_\star) = 7441/39434$.
α, β	6614/6615 and 73843/6615, coefficients in the upper bound on savings.
$e_{\mathrm{quarter}}, \bar{e}$	The gap $1 - h(1/4)$ and its rational upper bound 754889/4000000.

References

[1] *Coupled Reference-Process Bounds for Dense Adaptive Group Testing*, unpublished mathematical manuscript, October 25, 2026. Mathematical source for this exposition; all results needed here have been proved within the present document.

[2] M. C. Hu, F. K. Hwang, and J. K. Wang, A boundary problem for group testing, *SIAM Journal on Algebraic and Discrete Methods* **2**(2) (1981), 81–87. doi:10.1137/0602011.